

# JavaScript Objects (Part 1)

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## What is an Object?

### DEFINITION

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An **Object** is a collection of related data stored as **key-value pairs**.

Objects allow you to store multiple related values in a single variable.

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## Why Do We Use Objects?

Objects are useful when you want to store information about one thing.

For example:

- User Information
- Product Details
- Student Data
- Employee Information
- Car Details

Instead of creating many variables:

```
let name = "Rokon";  
let age = 23;  
let country = "Bangladesh";
```

We can use one object.

```
const user = {  
  name: "Rokon",  
  age: 23,  
  country: "Bangladesh"  
};
```

---

## Object Structure

---

```
const user = {  
  name: "Rokon",  
  age: 23,  
  country: "Bangladesh"  
};
```

Object

user

↓

{

name : "Rokon"

age : 23

country : "Bangladesh"

}

Here,

name

is called the **Key (Property)**

"Rokon"

is called the **Value**

---

## Creating Objects

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There are several ways to create objects.

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## Method 1 (Most Common)

```
const student = {  
  
  name: "Rokon",  
  
  age: 23,  
  
  cgpa: 3.85  
  
};  
  
console.log(student);
```

### OUTPUT

---

```
{  
  name: "Rokon",  
  age: 23,  
  cgpa: 3.85  
}
```

---

## Method 2

Create an empty object.

```
const person = {};  
  
person.name = "Rokon";  
person.age = 23;  
  
console.log(person);
```

## OUTPUT

---

```
{  
  name: "Rokon",  
  age: 23  
}
```

---

# Properties

---

## | Definition

A **Property** is a key-value pair inside an object.

---

# Example

```
const laptop = {  
  
  brand: "HP",  
  
  model: "EliteBook",  
  
  ram: "16GB"  
  
};
```

## PROPERTIES

| Property | Value     |
|----------|-----------|
| brand    | HP        |
| model    | EliteBook |
| ram      | 16GB      |

Another Example

```
const car = {  
  
  brand: "Toyota",  
  
  color: "Black",  
  
  price: 25000  
  
};
```

---

# Accessing Properties

---

There are two ways.

---

## 1. Dot Notation (Recommended)

```
const student = {  
  
  name: "Rokon",  
  
  age: 23  
  
};  
  
console.log(student.name);  
  
console.log(student.age);
```

## OUTPUT

---

```
Rokon  
23
```

## 2. Bracket Notation

```
console.log(student["name"]);  
  
console.log(student["age"]);
```

## OUTPUT

---

```
Rokon  
23
```

## When to Use Bracket Notation?

Useful when the property name is stored inside a variable.

```
const student = {  
  
  name: "Rokon",  
  
  age: 23  
  
};  
  
let key = "name";  
  
console.log(student[key]);
```

## OUTPUT

---

Rokon

---

# Updating Properties

---

## | Definition

You can change the value of an existing property.

---

## Example

```
const student = {  
  
  name: "Rokon",  
  
  age: 23  
  
};  
  
student.age = 24;  
  
console.log(student);
```

## OUTPUT

---

```
{  
  name: "Rokon",  
  age: 24  
}
```

---

## | Update Multiple Properties

```
const product = {  
  
  name: "Laptop",  
  
  price: 50000  
  
};  
  
product.price = 45000;  
  
product.name = "Gaming Laptop";  
  
console.log(product);
```

### OUTPUT

---

```
{  
  name: "Gaming Laptop",  
  price: 45000  
}
```

---

## Adding Properties

---

### | Definition

New properties can be added at any time.

---

## Example

```
const student = {  
  name: "Rokon"  
};  
  
student.age = 23;  
  
student.country = "Bangladesh";  
  
console.log(student);
```

## OUTPUT

---

```
{  
  name: "Rokon",  
  age: 23,  
  country: "Bangladesh"  
}
```

---

## Real Project Example

```
const user = {  
  
  name: "John"  
  
};  
  
user.email = "john@gmail.com";  
  
user.role = "Admin";  
  
console.log(user);
```

---

## Deleting Properties

### Definition

The `delete` keyword removes a property from an object.

### Syntax

```
delete object.property;
```

## Example

```
const student = {  
  
  name: "Rokon",  
  
  age: 23,  
  
  cgpa: 3.90  
  
};  
  
delete student.cgpa;  
  
console.log(student);
```

## OUTPUT

---

```
{  
  name: "Rokon",  
  age: 23  
}
```

---

## Another Example

```
const car = {  
  brand: "Toyota",  
  color: "Black",  
  price: 25000  
};  
  
delete car.price;  
  
console.log(car);
```

### OUTPUT

---

```
{  
  brand: "Toyota",  
  color: "Black"  
}
```

---

# CRUD Operations with Objects

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| Operation | Description            |
|-----------|------------------------|
| Create    | Create a new object    |
| Read      | Access properties      |
| Update    | Change property values |
| Delete    | Remove properties      |

Example

```
const student = {  
  name: "Rokon"  
};  
  
// Create  
student.age = 23;  
  
// Read  
console.log(student.name);  
  
// Update  
student.age = 24;  
  
// Delete  
delete student.name;  
  
console.log(student);
```

# Real Project Example

---

## User Profile

```
const user = {  
  
  name: "John",  
  
  email: "john@gmail.com",  
  
  isVerified: true  
  
};  
  
console.log(user.name);  
  
user.isVerified = false;  
  
user.phone = "01712345678";  
  
delete user.email;  
  
console.log(user);
```

---

## Best Practices

---

✓ Use `const` when creating objects.

```
const user = {};
```

---

- ✓ Use **dot notation** whenever possible.

```
user.name
```

---

- ✓ Use meaningful property names.

```
firstName
```

```
lastName
```

```
email
```

```
phone
```

---

## Common Mistakes

---

### | Accessing Wrong Property

```
const user = {  
  name: "Rokon"  
};  
  
console.log(user.age);
```

Output

```
undefined
```

---

## | Forgetting Quotes in Bracket Notation



```
user[name]
```



```
user["name"]
```

---

## | Deleting a Non-Existing Property

```
delete user.salary;
```

No error occurs, but nothing is deleted.

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# Interview Questions

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## | What is an Object?

An object is a collection of key-value pairs used to store related data.

---

## | What is a Property?

A property is a key-value pair inside an object.

---

## | How do you access object properties?

- Dot Notation

```
user.name
```

- Bracket Notation

```
user["name"]
```

---

## | How do you update a property?

```
user.age = 24;
```

---

## | How do you add a property?

```
user.country = "Bangladesh";
```

---

## | How do you delete a property?

```
delete user.age;
```

---

# Summary Table

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| Topic    | Description                             |
|----------|-----------------------------------------|
| Object   | Collection of key-value pairs           |
| Property | Key-value pair inside an object         |
| Create   | <code>{ }</code>                        |
| Access   | Dot / Bracket notation                  |
| Update   | <code>object.property = value</code>    |
| Add      | <code>object.newProperty = value</code> |
| Delete   | <code>delete object.property</code>     |
| CRUD     | Create, Read, Update, Delete            |

## JavaScript Objects (Part 2A)

---

This section covers:

- Object Methods
- this Keyword
- Real Project Examples
- Common Mistakes
- Interview Questions

---

## Object Methods

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### Definition

An **Object Method** is a function stored inside an object.

Methods allow an object to perform actions.

Unlike properties, methods contain executable code.

---

## Object Structure

```
const person = {  
  
  name: "Rokon",  
  
  age: 23,  
  
  greet: function () {  
  
    console.log("Hello!");  
  
  }  
  
};
```

Here,

| Item  | Type     |
|-------|----------|
| name  | Property |
| age   | Property |
| greet | Method   |

---

# Creating an Object Method

---

## Syntax

```
const object = {  
  
  methodName: function(){  
  
    // code  
  
  }  
  
};
```

## Example

```
const student = {  
  
  name: "Rokon",  
  
  greet: function(){  
  
    console.log("Welcome!");  
  
  }  
  
};  
  
student.greet();
```

## OUTPUT

---

Welcome!

---

# ES6 Short Method Syntax

---

JavaScript provides a shorter syntax for methods.

Instead of writing:

```
const user = {  
  
  greet: function(){  
  
    console.log("Hello");  
  
  }  
  
};
```

Write

```
const user = {  
  
  greet(){  
  
    console.log("Hello");  
  
  }  
  
};
```

Output

Hello

This is the preferred syntax in modern JavaScript.

---

# Methods Can Use Properties

---

```
const student = {  
  
  name: "Rokon",  
  
  greet(){  
  
    console.log("Hello " + this.name);  
  
  }  
  
};  
  
student.greet();
```

Output

```
Hello Rokon
```

---

## Returning Values

---

Methods can return values.

```
const calculator = {  
  
  add(){  
  
    return 20 + 30;  
  
  }  
  
};  
  
console.log(calculator.add());
```

Output

50

---

# Multiple Methods

---

```
const calculator = {  
  
  add(a,b){  
  
    return a+b;  
  
  },  
  
  subtract(a,b){  
  
    return a-b;  
  
  }  
  
};  
  
console.log(calculator.add(20,10));  
  
console.log(calculator.subtract(20,10));
```

Output

```
30
```

```
10
```

---

## Why Use Methods?

---

Methods group behavior with data.

Example

Instead of

```
let name = "Rokon";

function greet(){

  console.log("Hello");

}
```

Use

```
const user = {

  name: "Rokon",

  greet(){

    console.log("Hello");

  }

};
```

Everything related to the user stays inside one object.

---

# The this Keyword

---

## Definition

The **this** keyword refers to the object that is calling the method.

---

## Example

```
const student = {  
  
  name: "Rokon",  
  
  age: 23,  
  
  introduce(){  
  
    console.log(this.name);  
  
  }  
  
};  
  
student.introduce();
```

Output

```
Rokon
```

---

## | Explanation

```
student
```

```
↓
```

```
introduce()
```

```
↓
```

```
this
```

```
↓
```

```
student
```

Therefore,

```
this.name
```

means

```
student.name
```

---

# Access Multiple Properties

---

```
const student = {  
  
  name: "Rokon",  
  
  age: 23,  
  
  cgpa: 3.90,  
  
  details(){  
  
    console.log(this.name);  
  
    console.log(this.age);  
  
    console.log(this.cgpa);  
  
  }  
  
};  
  
student.details();
```

Output

```
Rokon  
  
23  
  
3.9
```

---

# Without this

---

```
const student = {  
  
  name: "Rokon",  
  
  greet(){  
  
    console.log(name);  
  
  }  
  
};  
  
student.greet();
```

Output

```
ReferenceError
```

Because JavaScript searches for a variable named `name`, not the object's property.

Correct

```
this.name
```

---

# this in Another Object

---

```
const car = {  
  
  brand: "Toyota",  
  
  model: "Corolla",  
  
  show(){  
  
    console.log(this.brand);  
  
    console.log(this.model);  
  
  }  
  
};  
  
car.show();
```

Output

```
Toyota  
  
Corolla
```

---

# Real Project Example 1

---

## | User Profile

```
const user = {  
  
  firstName: "MD",  
  
  lastName: "Rokon",  
  
  fullName(){  
  
    return this.firstName + " " + this.lastName;  
  
  }  
  
};  
  
console.log(user.fullName());
```

Output

```
MD Rokon
```

---

# Real Project Example 2

---

## | Shopping Cart

```
const product = {  
  
  name: "Laptop",  
  
  price: 50000,  
  
  discount: 5000,  
  
  finalPrice(){  
  
    return this.price - this.discount;  
  
  }  
  
};  
  
console.log(product.finalPrice());
```

Output

```
45000
```

---

# Real Project Example 3

---

## | Bank Account

```
const account = {  
  
  owner: "Rokon",  
  
  balance: 10000,  
  
  deposit(amount){  
  
    this.balance += amount;  
  
  }  
  
};  
  
account.deposit(5000);  
  
console.log(account.balance);
```

Output

```
15000
```

---

# Real Project Example 4

---

## | Student Result

```
const student = {  
  
  name: "Rokon",  
  
  marks: [80,90,95],  
  
  average(){  
  
    return (80+90+95)/3;  
  
  }  
  
};  
  
console.log(student.average());
```

Output

```
88.33333333333333
```

---

## Best Practices

- ✓ Use ES6 method syntax.

```
greet(){  
  
}
```

---

- ✓ Use `this` to access properties inside methods.

```
this.name
```

---

- ✓ Keep related data and functions inside one object.
- 

- ✓ Use meaningful method names.

## Examples

```
login()  
  
logout()  
  
calculateTotal()  
  
getFullName()  
  
addToCart()
```

---

# Common Mistakes

---

## | Forgetting Parentheses

Wrong

```
console.log(user.greet);
```

Output

```
[Function]
```

Correct

```
console.log(user.greet());
```

---

## | Forgetting this

Wrong

```
console.log(name);
```

Correct

```
console.log(this.name);
```

## Using Arrow Functions as Object Methods

Wrong

```
const user = {  
  name: "Rokon",  
  greet: () => {  
    console.log(this.name);  
  }  
};
```

`this` inside an arrow function does **not** refer to the object.

Correct

```
const user = {  
  
  name: "Rokon",  
  
  greet(){  
  
    console.log(this.name);  
  
  }  
  
};
```

---

## | Typo in Property Names

Wrong

```
this.fullname
```

Correct

```
this.fullName
```

JavaScript property names are case-sensitive.

---

# Interview Questions

---

## | What is an Object Method?

A method is a function stored inside an object.

---

## | What is the difference between a Property and a Method?

| Property                   | Method                        |
|----------------------------|-------------------------------|
| Stores data                | Stores a function             |
| Example: <code>name</code> | Example: <code>greet()</code> |

---

## | What does `this` refer to?

`this` refers to the object that calls the method.

---

## | Why should you use `this`?

It allows a method to access the object's own properties.

---

## | Can an object have multiple methods?

Yes.

```
const calculator = {  
  
  add() {},  
  
  subtract() {},  
  
  multiply() {},  
  
  divide() {}  
  
};
```

## | Which syntax is preferred for object methods?

```
greet(){  
  
}
```

instead of

```
greet: function(){  
  
}
```

## Can methods return values?

Yes.

```
square(n){  
  
    return n*n;  
  
}
```

## Summary Table

| Topic             | Description                            |
|-------------------|----------------------------------------|
| Method            | Function inside an object              |
| ES6 Method Syntax | <code>greet(){}</code>                 |
| this              | Refers to the current object           |
| Access Property   | <code>this.property</code>             |
| Return Value      | <code>return</code> statement          |
| Call Method       | <code>object.method()</code>           |
| Best Practice     | Use ES6 methods with <code>this</code> |